

RipSetCocoaCNCH12: Labeled Dataset for Ripeness Stage Detection, Semantic and Instance Segmentation of Cocoa Pods

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Abstract: Fruit counting and ripeness detection are computer vision applications that have gained strength in recent years due to the advancement of new algorithms, especially those based on artificial neural networks (ANNs), better known as deep learning. In agriculture, those algorithms capable of fruit counting, including information about their ripeness, are mainly applied to make production forecasts or plan different activities such as fertilization or crop harvest. This paper presents the RipSetCocoaCNCH12 dataset of cocoa pods labeled at four different ripeness stages: stage 1 (0–2 months), stage 2 (2–4 months), stage 3 (4–6 months), and harvest stage (>6 months). An additional class was also included for pods aborted by plants in the early stage of development. A total of 4116 images were labeled to train algorithms that mainly perform semantic and instance segmentation. The labeling was carried out with CVAT (Computer Vision Annotation Tool). The dataset, therefore, includes labeling in two formats: COCO 1.0 and segmentation mask 1.1. The images were taken with different mobile devices (smartphones), in field conditions, during the harvest season at different times of the day, which could allow the algorithms to be trained with data that includes many variations in lighting, colors, textures, and sizes of the cocoa pods. As far as we know, this is the first openly available dataset for cocoa pod detection with semantic segmentation for five classes, 4116 images, and 7917 instances, comprising RGB images and two different formats for labels. With the publication of this dataset, we expect that researchers in smart farming, especially in cocoa cultivation, can benefit from the quantity and variety of images it contains.

Keywords: cocoa pods detection; ripeness stage detection; semantic segmentation; smart farming



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1. Introduction

The application of precision agriculture strategies in cocoa crops continues to encounter various challenges that need to be addressed. These challenges primarily involve issues related to the poor quality of existing data and the acquisition of new data necessary for the application of advanced precision agriculture techniques [1].

One of the main challenges is to identify different stages of ripeness of the cocoa pods since this type of crop has a wide number of varieties, and all of them can show different textures and color characteristics in their maturation process [2].

Detecting ripeness stages in cocoa pods is critical in determining two relevant factors in any crop: effectively planning the optimal timing of harvest and accurately predicting production volumes. Unfortunately, the adequate maturity to harvest is not always homogeneous, affecting the fermentation process necessary to obtain good chocolate quality [3].

The following different techniques have been tested for ripeness-stages detection in cocoa pods:

- acoustic signals [2,4],
- determination of metabolic profiles through biochemical markers [5], and

- laser techniques with backscattered images [6].

However, these techniques are unrealistic when implemented in the field with real conditions, since the devices for capturing sound data, laser images, spectrometry, or bio-chemical markers require expensive devices that are not within the reach of the farmers.

On the other hand, artificial intelligence techniques based on artificial neural networks (ANNs), better known as deep learning, are increasingly used [7–9].

The precision and robustness of deep learning models depend on the quality and quantity of the training data, as they are crucial factors that contribute to the variability of the phenomenon under study [10].

Moreover, the increasing prevalence of smartphones among farmers for their daily activities simplifies the process of capturing images, eliminating the necessity of investing in costly equipment and specialized management for data capture.

Unfortunately, the community engaged in applied research using deep learning techniques to detect ripeness stages in cocoa pods faces a scarcity of image datasets for most varieties. In addition, the available public datasets offer only a limited number of images for training deep learning models [8,11].

To help the community that performs applied research for developing deep learning solutions to detect ripeness stages in cocoa pods, we propose the RipSetCocoaC-NCH12 dataset, which consists of 4116 images taken with different types of smartphones labeled for semantic segmentation. Having several stages of ripeness is a feature that will allow researchers to train machine learning algorithms that classify more than two classes: mature and immature. These features will allow the scientific community interested in these applications to train more robust and accurate deep learning models.

The RipSetCocoaCNCH12 dataset will be important for the training of machine learning algorithms that seek to detect different ripeness stages in cocoa crops of the CNCH12 variety and to make inventories of pods.

2. RipSetCocoaCNCH12 Dataset

2.1. Description

CACAO CNCH12, developed by “Compañía Nacional de Chocolates”, is the cocoa variety in the dataset. The images were collected at the “Compañía Nacional de Chocolates” farm, located in the municipality of Támesis, department of Antioquia—Colombia (5°43′02″ N–75°41′25″ W). The average height above sea level in the farm is approximately 1100 m. The dataset was created between 1 December 2022 and 17 February 2023, the primary cocoa harvest season in the study area.

The average ripening period for a cocoa pod typically spans six to seven months, although slight variations may occur based on the specific agronomic and climatic conditions of the crop. The ripeness stages were defined in ranges of two months due to the key physical and chemical differences of the cocoa pods according to the agronomists of the “Compañía Nacional de Chocolates” company. The stages are defined based on the duration in months, starting from pollination of the flowers to the optimal time for harvesting the pod. The sequential progression of cocoa pods during the ripening process, from 0 to 6 months, is illustrated in Figure 1.



Figure 1. Ripeness process in a sequence of cocoa pods.

The images of cocoa pods were divided into five classes (Table 1). They were divided into four classes according to their ripeness stage in months: Class 1 (0–2 months), Class 2 (2–4 months), Class 3 (4–6 months), and Class 4 (>6 months) (Figure 2). Additionally, there is a fifth class known as “abortions” that does not fall under any of the ripeness stages (Class A). Abortions are cocoa pods that start their growth process but die from various causes associated with attacks by pests or diseases or even due to physiological problems of the plant (Figure 3).

Table 1. Number and names of instances per class.

Class	Class Name	Instances
C1	Stage 1 (0–2 months)	3278
C2	Stage 2 (2–4 months)	1688
C3	Stage 3 (4–6 months)	1519
C4	For harvest (>6 months)	1169
CA	Abortions	263
Total		7917



Figure 2. Ripeness stages: (a) 0–2 months (C1); (b) 2–4 months (C2); (c) 4–6 months (C3); (d) >6 months (C4).



Figure 3. Examples of several types of abortions (CA).

The dataset contains two folders: the first contains the annotations in COCO 1.0 format, and the second contains the images in segmentation mask 1.1 format. In each of these folders, the images are divided into subfolders named with the main class they contain; an image can contain several instances of different classes, but the images in each folder are dominated by one of the classes. The distribution of instances in each folder can be seen below in Figure 4.

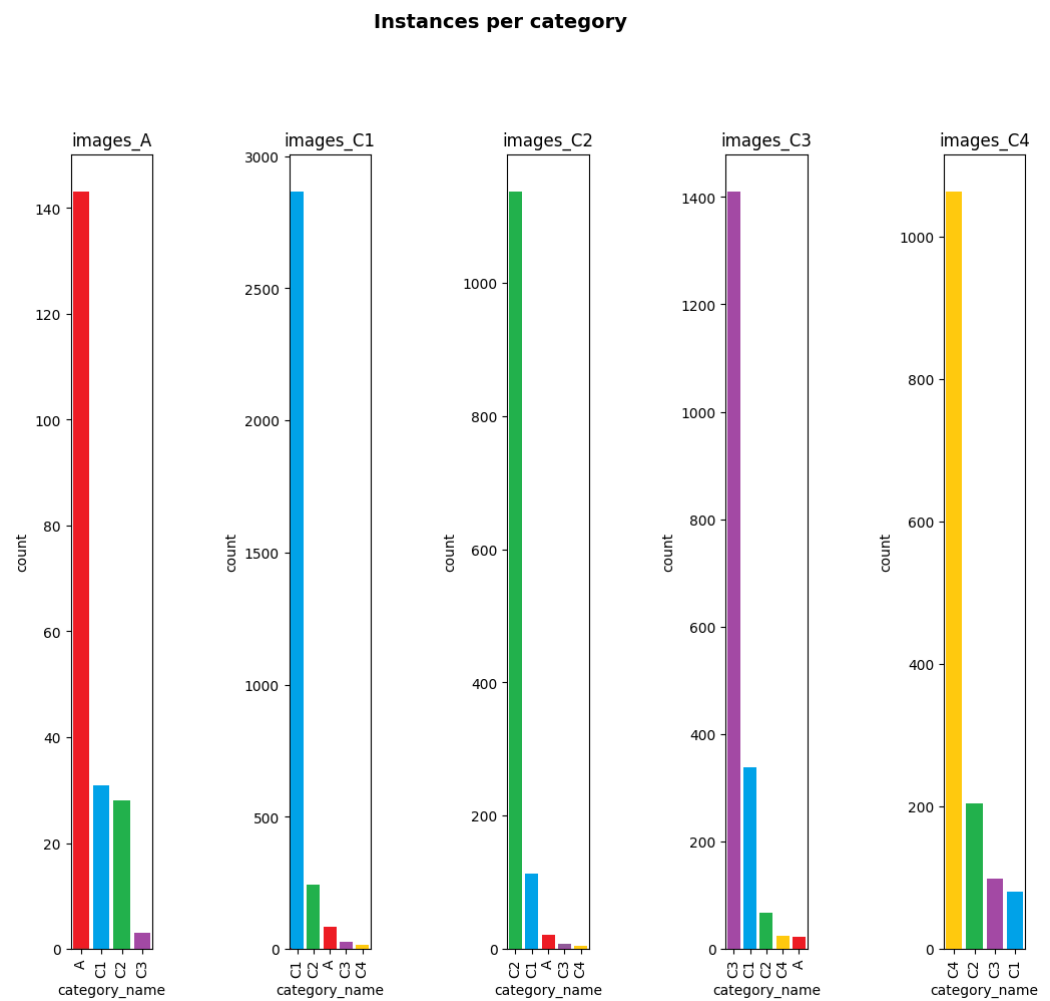


Figure 4. Distribution of the instances for each image folder (*y*-axes differ between the frames).

2.2. Quantitative Measure to Differentiate Cocoa Classes

The ripening process of fruit involves a sequence of physiological changes to become ready for consumption or processing. The fruit grows, accumulating essential nutrients and water, while noticeable transformations in color, texture, and composition signify its ripeness.

A widely used way to measure the state of maturity of a fruit quantitatively at different stages is to calculate the internal sugar content by measuring Brix degrees [12–15]. To have a quantitative measure that would confirm the difference between ripeness stages, the Brix degrees were measured in more than 35 cocoa pods for each class in the four ripeness stages (C1 to C4). The results are presented in Table 2.

Table 2. Number of samples and average Brix degrees for the ripeness stages.

Class	Number of Samples by Class Selected to Measure Degrees Brix	Average Brix Degrees Measured (μ_j) (°Bx)
C1	39	5.3
C2	45	6.6
C3	38	8.7
C4	40	16.6

An ANOVA test was performed to check for a significant difference between the different classes, according to their measure of Brix degrees. The results can be seen below in Table 3.

Table 3. ANOVA table for Brix degrees in the four different ripeness stages.

Source of Variation	Sum of Squares	df *	Mean Square	F **	p Value
Between groups	2955.78	3	985.26 (VBG)	305.72	$1.12e^{-63}$
Within groups	483.41	150	3.22 (VWG)		
Total	3439.20	153			

* Degrees of freedom. ** F = variance between groups/variance within groups = VBG/VWG.

Null hypothesis : μ_j are equal

Alternative hypothesis : μ_j are not equal

According to the results of the F and p-value, the null hypothesis is rejected. Therefore, there is a significant difference in Brix degrees among classes, which confirms the accuracy of dividing cocoa pods into the four proposed classes for the stages of ripeness.

Every image is 3000×3000 px in JPEG format, with 8 bits. The image files were named with the date and time of capture. Figure 5 is an example of the images corresponding to the four ripeness stages.

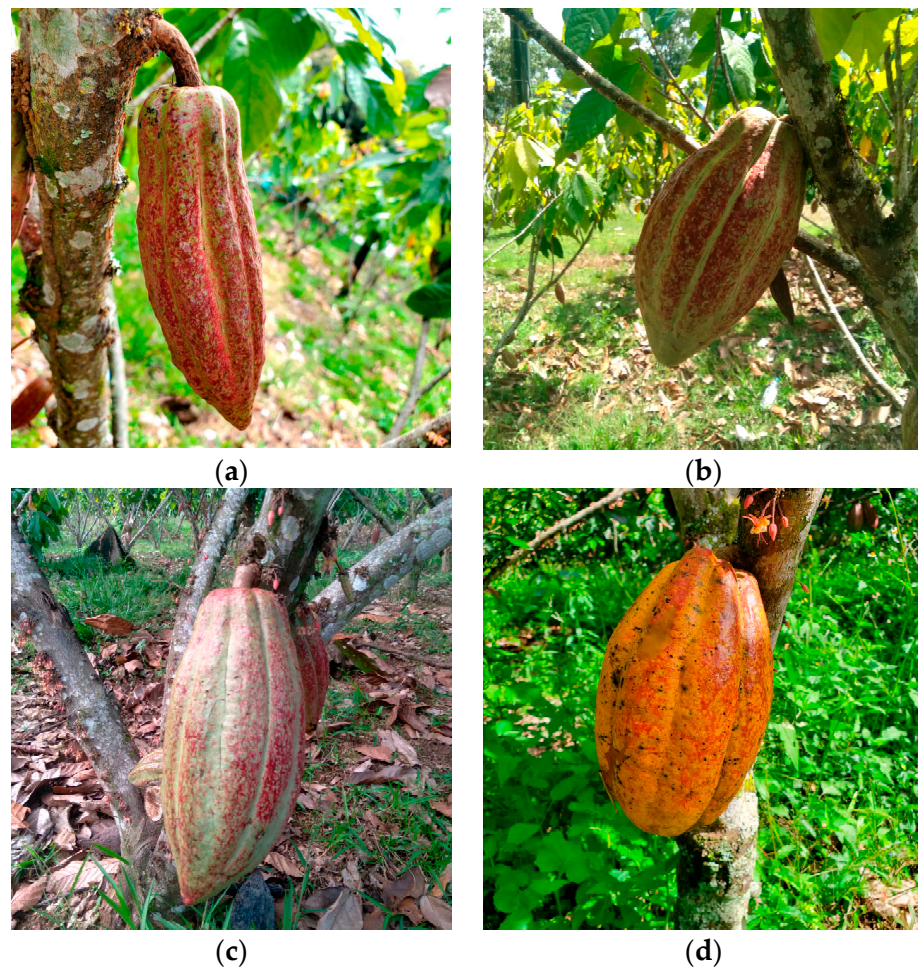


Figure 5. Dataset examples of the ripeness stages: (a) Class 1; (b) Class 2; (c) Class 3; (d) Class 4.

Table 4 below shows a summary of the RipSetCocoaCNCH12 dataset.

Table 4. The RipSetCocoaCNCH12 specifications.

Item	Description
Field of application	Object detection—smart farming
Data acquisition	Smartphone devices
Method of annotation	Manually with CVAT (Computer Vision Annotation Tool)
Number of classes	5: stage 1 (0–2 months), stage 2 (2–4 months), stage 3 (4–6 months), for harvest (>6 months), and abortions
Number of images	4116
Number of instances	7917
Data collected by	Authors of this paper
Years of collection	2022–2023
Vertical resolution	96 dpi
Horizontal resolution	96 dpi
Dataset size	27 GB
Image format	.JPG
Image size	3000 × 3000 px
Annotation formats	COCO 1.0 and segmentation mask 1.1

3. Methods

Nowadays, smartphones have become ubiquitous. In even the most remote rural areas, smartphones have become the main communication technology due to their low costs and portability. These devices can also give farmers the ability to collect image data. Therefore, in this work, the images were captured with smartphones to have a dataset as similar as possible to real conditions.

3.1. Image Data Acquisition

Five devices from some of the leading manufacturers were selected for this work. To ensure significant variability in the types of images captured and enrich the dataset, multiple devices were chosen. The technical specifications of used smartphones can be seen below in Table 5.

Table 5. Technical specifications of the smartphone cameras used to capture the dataset images.

Smartphone	Camera Specifications
Samsung Galaxy A01	Dual rear camera consisting of a 13-megapixel f/2.0 main sensor and a 2-megapixel f/2.4 depth sensor.
Samsung Galaxy Note 10	Triple camera composed of an ultra-wide angle: 16 MP, f/2.2, 123°; a wide angle: 12 MP, AF, f/1.5–2.4; and a phone Camera: 12 MP, f/2.1.
iPhone SE 2020	Single camera. 12 MP wide-angle camera, f/1.8 aperture.
LG G5	Dual camera. 16 MP main camera and f/1.8 aperture. 8MP secondary super-wide-angle camera with f/2.4 aperture.
Motorola G9 plus	Quadruple camera. Main camera: 64 MP sensor, f/1.8 aperture and phase detection focus. Ultra-angular: 8 MP sensor, f/2.2 aperture. Macro: 2 MP sensor and f/2.2 aperture. Depth: 2 MP sensor and f/2.2 aperture.

The strategy for capturing images involved zigzag paths in the field enabling access to each crop tree. During each pass, a person took images of a single class to allow easier classification in the folders.

Between one and four images of each cocoa pod were taken from different angles to obtain as many samples as possible (Figure 6).

The images were taken between 8:00 a.m. and 4:00 p.m. First, the size format for the capture was adjusted on all smartphones to a 1:1 ratio, and then resizing was applied to them using a script in the Python language with Pillow (Python Imaging Library), giving

them a final size of 3000×3000 px. The original images had sizes in the range from 3072×3072 to 4096×4096 px.



Figure 6. Image capture process for one cocoa pod from different angles.

3.2. Brix Degrees Data Acquisition

Some pods were selected to measure the Brix degrees of the internal sugar content, as mentioned in Section 2.1. First, the pods chosen for samples were perforated with a drill. Then, the sample was extracted, which was later placed in a handheld refractometer, and finally, the data were recorded manually. Images of this process can be seen below in Figure 7.

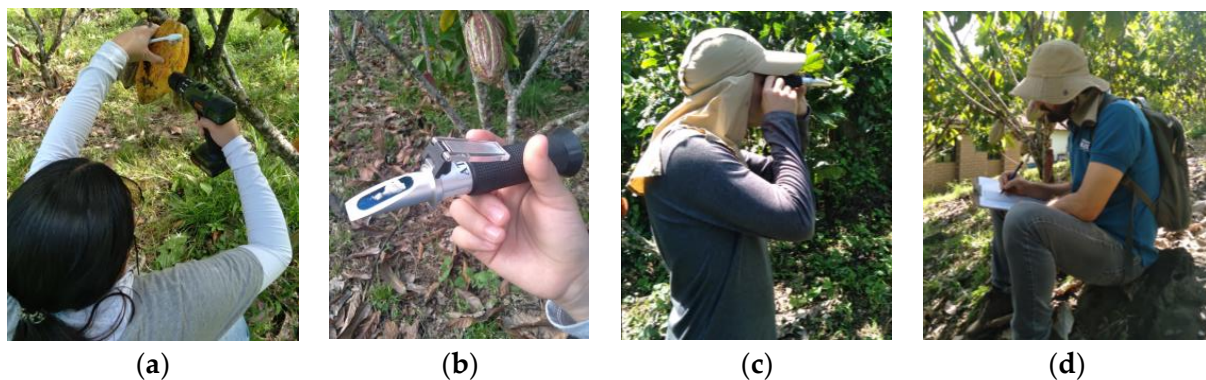


Figure 7. Process of sampling to measure Brix degrees: (a) perforation of the cocoa pod with a drill, (b) placement of the sample in a handheld spectrometer, (c) reading of the Brix degree measurement, and (d) measurement recording.

3.3. Data Annotation

The tool used for labeling images was CVAT (Computer Vision Annotation Tool) [16], which allows for different techniques. The technique used for this work was polygon labeling to obtain a semantic segmentation of the classes (Figure 8).

The dataset contains labels in two alternative formats: (1) COCO 1.0, which has files in the format (*.json) for detection using bounding boxes and polygons, and (2) segmentation mask 1.1, which contains separate folders for semantic segmentation and instance segmentation. Examples of these masks can be seen in Figures 9 and 10.



Figure 8. Examples of labeling of cocoa pods with CVAT.

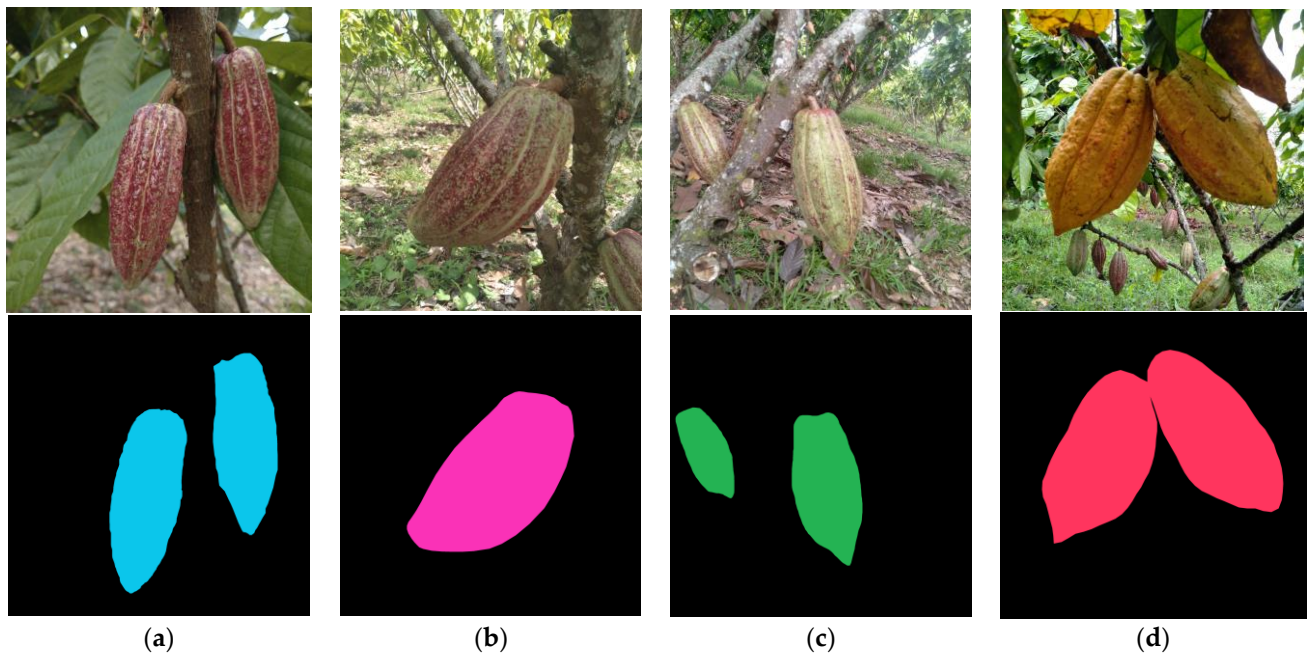


Figure 9. Examples of masks for semantic segmentation: (a) C1; (b) C2; (c) C3; (d) C4.

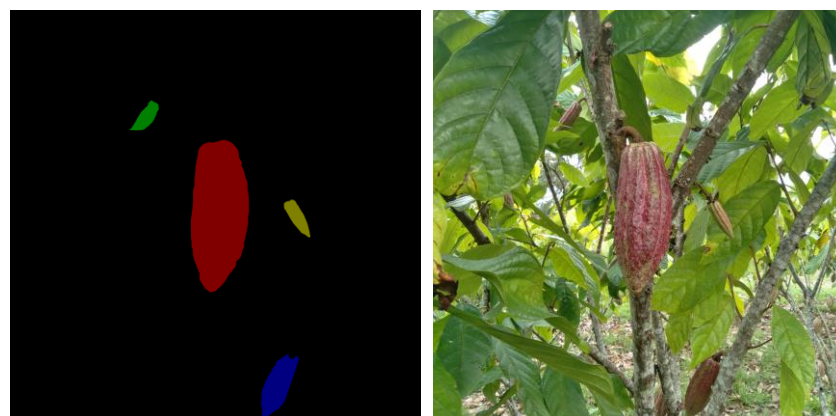


Figure 10. Example of masks for instance segmentation: Class 1.

4. Limitations

The RipSetCocoaCNCH12 dataset does not include classes of cocoa pods to discard. In future work, diseases and rotten pods may be included. Additionally, more data should be collected on other different cocoa varieties.

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